



Real-Time News Classification and Summarization Using Machine Learning, Deep Learning, and Large Language Models

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Abstract –The Increasing traffic of online news and live data has developed a new requirement of algorithms capable of organizing and summarizing our daily media reports in real time. This project makes an attempt to create an end-to-end news classification and summarization system using classical Machine Learning (ML), Deep Learning (DL), and Large language Models (LLM's) like GPT, Llama etc. This system is trained using the BBC News dataset, which consists of labelled articles across a variety of categories. Classical ML models, including Logistic Regression, Naive Bayes, Support Vector Machine, and Random Forest, are integrated and compared with Deep Learning (Neural-Network) models such as Long Short-Term Memory (LSTM) networks to derive classification performance.

The model selection is done according to the classification metrics such as Accuracy, Precision, Recall, and F1-Score. In order to improve real-world application, the trained classifier is used in a real-time pipeline that extracts ongoing news articles from external news APIs. The system is designed in such a manner that it shall first preprocess the news article text fetched via APIs and then it shall classify those articles into some predefined categories. Along with this a transformer-based LLM shall also be utilised in order to generate clean and concise summary of the fetched news articles so the content is easily accessible to the users.

This project is designed to create a single system that solves a real life NLP issue and improves readability and accessibility of news articles.

Keywords: News Classification, Text Processing, Machine Learning, Deep Learning, Natural Language Processing (NLP), LSTM Networks, Transformer Models

1 Introduction

The quantity of online news content has increased substantially due to the development of several digital media platforms [1]. Millions of news articles are being published worldwide across a variety of fields such as business, sports, entertainment and more. This enhances accessibility and user experience. However, it also makes it really tough to properly organize and categorize content.

Manual categorization of news articles is really tough, error-prone, and is not scalable for real-time situations. Because of this, automated text classification systems have become an important component of online news platforms. Machine Learning and Deep Learning techniques have shown signs of proven success in Natural Language Processing (NLP) tasks, including document categorization and text summarization [2]. Classical ML models such as Logistic Regression, SVC's, Gaussian Naïve Bayes are proven to perform well for structured text classification tasks such as spam detection or sentiment classification. However, advanced Deep Learning models such as LSTM/GRU's and Transformer models show better results for sequential data such as text or time series due to their ability to understand context and retain information over previous time steps.

Apart from classification, machine based text summarization has gained importance because of the increase in demand for quicker and efficient consumption of information. Encoder-Decoder based Transformer models like GPT or Llama are found to be really efficient for producing clean, concise and meaningful summaries of lengthy documents such as news articles.

Despite these advancements in Machine Learning, there is still a need for a system that can accurately classify news, fetch real-time articles, and produce the summaries of those articles within a single unified framework [3]. This project aims to develop an end-to-end smart news processing system that utilizes traditional models, Neural Network based models, and LLMs to classify and generate summaries of the fetched news articles in real time.

Contribution

The main contributions of this project are as follows:

1. Developing an end-to-end intelligent system that integrates news article classification and summarization into a single pipeline.
2. Comparing and analysing multiple Statistical and Deep Learning models for news classification and analysing the most effective approach.
3. Implementing a real-time news collection pipeline using external news APIs to dynamically fetch articles.
4. Integrating an LLM to generate concise and meaningful summaries of classified news articles.
5. Deployment-oriented design that bridges the gap between theoretical research models and practical real-world applications.

2 Problem Statement

The rapid surge and growth of e-news on various multiple platforms, has led significant challenges in organizing, and consuming the feed properly. Every day, millions of news articles are published across different domains such as business, politics, sports, tech etc. This massive number of unstructured articles makes it hard for users to quickly identify relevant information. Physical categorization of news articles consumes lots of time, and is not scalable, especially in real-time situations where news is continuously generated.

Although many news platforms in today's period provide classified content, their categorization methods often rely on stationary or semi-automatic processes, that may lack accuracy, and real-time handling capabilities. Also, existing systems mostly focus on classification and don't provide summarization of articles. As a result, users must read entire articles to understand the key points of information, which reduces effectiveness and impacts user experience, especially when dealing with large amounts of content.

In addition, most implementations focus on only either Machine Learning or Deep Learning models using the stationary datasets. Incredibly Limited work has been done to fit in real-time news extracting, automatic classification, and summarization into a unified system. The absence of such integrated solutions restricts the real-life applicability of news categorization models.

Therefore, there is a serious requirement to develop a smart, scalable, and automated real-time news classification and summarization system that can build an intelligent system that automatically sort news articles into clear, organized categories using Machine Learning or Deep Learning tactics, create a pipeline to pull in real-time news articles and generate short and meaningful summaries using Large Language Models (LLMs).

This system is designed for classifying news, fetching news articles in real time, and summarize news articles using LLM into a unified framework. The system aims to improve the efficacy and accuracy in news classification while also enhancing the user experience by generating clear summaries of long articles. The system also demonstrates that how classical ML, Neural Networks, and LLM's can be used together in order to solve real-life challenges.

3 Objectives

The main goal of this project is to design and evaluate a smart system for classifying and summarizing news in real time. This will be done using ML, DL and LLM's. This system aims to sort the news articles into predefined categories and generate a summary of the articles. This will improve how people access and use news content in their daily lives. The concerns handled by this project are as follows:

1. **Collecting, Preprocessing and analysing BBC News dataset:** The articles will be cleaned, tokenization will be performed along with techniques like stop-word removal (removing frequently occurring words), lemmatization, and normalization to further improve quality of the data. Feature extraction techniques such as TF-IDF and BOW shall be used to convert textual data into their numerical representations which makes it suitable for training ML models.
2. **Implementing and training multiple traditional Machine Learning models:** Statistical ML models including Logistic Regression, Naïve Bayes, ensemble models etc. shall be trained simultaneously on dataset to classify the news articles into predefined categories. The model performance shall be compared using classification metrics such as Precision, Recall, F1-Score etc that will measure their effectiveness in automated news classification.
3. **Developing Deep Learning-based models:** Neural Network based models such as LSTM's / Bi-LSTM's or GRU's shall be trained on the dataset which would help capture contextual nuances in the articles. A Self Attention Layer may also be incorporated for better results. The performance of these models shall then be compared with traditional ML models.
4. **Developing automated real-time news system:** A news fetching mechanism shall be utilised to get news articles via APIs such as News API. The system will extract relevant information such as article link, article text, title etc. which would be beneficial for accurate classification and summarization.
5. **Implementing a real-time news classification pipeline:** The fetched articles shall be pre-processed and passed through the trained classifier to automatically categorize it.
6. **Integrating LLM-based summarization module:** Transformer based LLM's such as Llama/Gemini shall be used for creating meaningful summaries of lengthy news articles, which would further enhance user experience.
7. **Developing an intelligent article processing system:** This system shall integrate real-time news collection, data preprocessing, model training, classification, and summarization into a single system, which shall showcase the use of NLP in solving some real-world problems.

4 Literature Review

One of the core applications of NLP is text classification, of which thorough research has been done and is still being done using various traditional ML models like Naïve Bayes, Neural Network based models like LSTM and advance transformers based models such as BERT.

News articles classification has also gained significant attention as millions of news articles are being published worldwide on various digital media platforms across a variety of fields including business, sports, tech, and more. Although such huge information across the internet enhances the accessibility and user experience but it also makes it extremely difficult to organize, categorize and consume relevant news properly.

Statistical machine learning models like Logistic Regression, Support Vector Machines (SVM) and probabilistic models like Naïve Bayes were primarily used during early research in text classification approaches. These models worked by converting the textual data into their numeric/vector representation using feature extraction techniques like Bag of Words (BOW) and Term Frequency-Inverse Document Frequency (TF-IDF). Among these models, Naïve Bayes, proposed by McCallum and Nigam (1998) was very popular and well known for text classification tasks due to its simplicity, ease of use and faster performance on large-scale text datasets especially where independence of features is assumed.

Simultaneously, Support Vector Machines (SVC) also attracted a lot of attention after Joachims (1998) showed and demonstrated how well they performed in text categorization tasks. These researches have demonstrated that although Naïve Bayes is faster and computationally efficient, however, SVM frequently outperform conventional probabilistic models in terms of accuracy. The main limitation of these model's is that they heavily rely on manual feature engineering and processing and also lack the ability to understand and capture semantic meaning and relationships within the text. Therefore, performance of these models on complex real-world textual data such as emotion classification or text classification may be limited.

As machine learning techniques advanced, text classification has also been benefited with the use of ensemble techniques and models such as bagging techniques like Random Forest. By combining multiple decision trees, and handling the challenge of overfitting, it has improved text classification performance. Studies have shown that these ensemble methods can produce reliable results on a variety of datasets.

The performance of text classification systems has greatly been benefitted with the advent of deep learning and neural network based models. LSTM, which was first presented by Hochreiter and Schmid Huber (1997), is a special type of recurrent neural network that is intended to handle sequential data like text while maintaining long term dependencies. These networks use word embeddings which are particularly good at understanding contextual relationship in text and hence they are very suitable and effective for tasks like text classification, emotion detection, time series analysis etc.

Kim (2014) made additional progress by introducing Convolutional Neural Networks (CNN) for sentence classification. These models have shown promising results in classification tasks. They are designed in such a manner that they automatically learn hierarchical feature representations from text which reduces the need of manual feature engineering to certain extent.

The field of Natural Language Processing (NLP) has been majorly transformed more recently with the introduction of transformer-based models. Bidirectional Encoder Representations from Transformers (BERT), introduced by Devlin et al. (2018), utilizes an encoder only architecture along with a bidirectional self-attention mechanism which helps it capture contextual relationships and better understand textual data. It produces state-of-art results in a variety of NLP tasks such as text classification, summarization, generating contextual embeddings etc.

Apart from classification, automatic text classification has also become also emerged as an important field in research. Traditionally, extractive summarization techniques were used which relied on choosing key sentences based on some statistical features. However, the latest advancements in LLMs such as GPT-based models or decoder-only models, have enabled abstractive summarization, wherein the model produces summaries that resemble those of a human by comprehending and understanding the semantic meaning of text.

A number of studies have also investigated ML based real time news classification. Pipelines for data collection, preprocessing, classification, and deployment are usually integrated in these systems. Nevertheless, a majority of implementations do not use LLMs for real time abstractive summarization of fetched news articles and instead focus primarily on building models for classification purpose.

A comparative analysis of various ML, and DL models was conducted by Karaman et al. (2023) [4] to evaluate the performance of these models on news classification. The BBC News dataset was used for this purpose. In this, GloVe embeddings were utilised for feature extraction purpose. The authors used various approaches such as SVM, LSTM and a CNN + LSTM architecture. The experiments showed that SVM achieved the best accuracy score of 98 %, compared to other approaches like LSTM and hybrid CNN and LSTM which achieved 97 % and 96 % accuracy respectively. This proves that classical ML models can still perform better than complex deep learning based models and achieve higher accuracy when used with simpler or smaller datasets and when proper feature engineering is performed.

In another study, Salman and Obaida (2021) [5] concentrated on news classification using a variety of Naïve Bayes Multinomial classifier, including, Gaussian, Bernoulli and Complement Naïve Bayes. In the proposed work, The authors demonstrated a much simpler pipeline which included standard text preprocessing techniques such as tokenization, stop word removal followed by feature extraction and model training and classification. TF-IDF technique was applied for feature extraction. In addition to this, the performance of these classifiers was assessed using standard classification performance metrics such as Accuracy, Precision, Recall, and F-1 Score. The experimental results of this study showed that Complement Naïve Bayes performed the best when compared to others and achieved highest accuracy of 97.6 %. This also proved the effectiveness of probabilistic ML models in text classification tasks when optimized properly along with suitable feature extraction techniques.

A separate system of classification as well as summarization was proposed by Dutt and Smalley in 2020 [6]. In this work the authors used various statistical machine learning models including SVM, Naïve Bayes, Logistic Regression and Random Forest for the purpose of news classification. The researchers focussed on the importance of feature preprocessing and utilized Bag of Words (BoW) representation for training each of the models. The trained models were evaluated based on accuracy and among all the evaluated models, SVM achieved the highest classification accuracy of 98.697%, followed by Logistic Regression at 98.293% and Naïve Bayes at 97.078 % while the Random Forest model achieved a significantly lower accuracy of only about 83%. This highlighted the challenge of overfitting and sensitivity of ensemble learning based models to sparse and high dimensional feature spaces.

Apart from classification techniques, the researchers also implemented a Text Rank algorithm for performing extractive summarization of the articles text in the dataset. It is a method of selecting most important sentences from a given piece of text based on statistical ranking techniques. This work effectively showed the use of ML models in classifying news articles as well Text Rank for summarising the news articles.

In a research articles (2014) [7], the researchers carried out a analysis of different supervised algorithms for news classification purpose. The researcher implemented a total of four different classifiers including Naïve Bayes, Random Forest, KNN and SVM to classify the news articles into pre-defined categories. The result showed that all the models performed well, however the SVM model showed better accuracy while KNN showed a relatively lower accuracy on the test dataset.

To summarize the findings, existing studies show that classical ML models are proven to be highly effective on this dataset when compared to advance neural network based deep learning models. This could be largely due to the structured and relatively simpler nature of the dataset where simpler techniques like TF-IDF or BoW are sufficient to express text as their numerical representations.

5 Research Gaps

Although several advancements and studies have been done in classification of news articles using Natural Language Processing. However there still exists some considerable gaps and limitations of the current researches.

There have been limited studies in investigating the potential of sequential neural networks like GRUs and advanced model combination such as Bi-LSTM architectures that can capture context of text from both ends for news classification purpose. Furthermore, attention mechanisms which can be used in combination with LSTMs to capture deep semantic meaning and contextual understanding of text have not been sufficiently and appropriately investigated for their application in classification of news articles.

Existing studies primarily focus on offline news classification wherein the model is trained on a static dataset and also tested on the same portion of that dataset, which sometimes makes them less applicable for real world use case and reduces their efficiency. Also, most of the methods presented in researches mostly focus either on static classification or summarization individually. Almost none of the studies have tried incorporating both into a single system for real world applicability. There is still a lack of system that could fetch live news from certain API's such as News API, automatically preprocess the articles and classify them into categories while also summarizing the text using an LLM.

To briefly summarize, the existing researches and approaches mainly rely on pre-processed static datasets and further cover the issues of real time article ingestion and processing. Limited work has been conducted in combining ML models and LLMs into a single framework.

Therefore, this again proves the need of an end-to-end integrated solution to effectively and reliably consume digital news using Natural Language Processing Techniques.

6 Proposed Methodology

This section covers the ideology and the proposed approach for the development of a real time news classification and summarization system. This methodology focusses on the integration of ML, DL, and Large Language Models (LLMs) together into a single unified system in order to solve a real life NLP issue.

The proposed system has been envisioned as a comprehensive and end to end pipeline for real time news classification as well as article summarization. It integrates various essential components which are essential for news extraction, article preprocessing, feature extraction, model training, classification etc. into a single comprehensive framework.

The best performing model, after comparison shall only be used and deployed for classification of incoming news articles. Also, an LLM shall also be integrated within the system in order to generate readable and concise summaries of the articles which would help enhance user experience.

6.1 Dataset collection

The dataset that shall be used in this study is collected from the BBC News Dataset, which is sourced from BBC Website and Kaggle. This dataset shall be used for the purpose of training as well as validating the models. It contains labelled textual data of news articles which is highly suitable for classification tasks in NLP.

In addition to this static data, several news articles shall be collected from some external News APIs, simulating real-world scenarios. These fetched articles will then be dynamically classified in real time by the trained model.

6.2 Data Preprocessing

This raw dataset cannot be directly used for training the machine learning models. Therefore, prior to training the models, certain steps shall be followed for preprocessing the dataset. The steps that shall be included are as follows:

- Lowercasing the news articles to ensure uniformity in the text, and at the same time reducing vocabulary size.
- Removing any trailing spaces from the text.
- Removing punctuations, special characters, stop words and noise from the dataset (while training traditional models)
- Performing tokenization on the sentences in order to convert the text into words, which is essential for training models.
- Performing lemmatization on the tokens in order to convert each token to their base form. Eg: beautifully and beautiful both gets converted into beautiful.
- Performing tokenization for DL models (converting text into numbers) and performing padding to enable equal length for each sentence which is ideal for training neural network based models.
- Converting the classes or categories into numbers using Label Encoding so that the models can understand them.

6.3 Feature Extraction

After the dataset is pre-processed, the text data shall be converted into their vector representation using common feature extraction techniques such as Bag Of Words as well as TF-IDF. The purpose of these techniques is that they transform text into numerical representations which is crucial for training machine learning models as these models don't directly process text.

In the case of the Bag Of Words technique, the text is converted into vectors, wherein the values in the vector represents the frequency of words in that document. This approach however, ignores the word order and only focuses on word count, this makes it computationally less expensive but this may affect the classification results. Let $V(\text{Vocabulary}) = \{w_1, w_2, w_3 \dots w_n\}$, Then a document or sentence can be represented as :

BoW = $[f_1, f_2, \dots, f_n]$, where f_i is the number of times that word occurs in that particular document.

TF-IDF technique is an improvement over this since it takes into account the word importance across the corpus. It is designed in such a manner that it reduces the effect of commonly occurring words in a sentence and increases the weightage of words that occur rarely.

$TF(t, d) = (\text{Number of times a term } t \text{ occurs in document } d) / (\text{Total terms in document } d)$

$IDF(t) = \log(N/n_t)$, where N is the number of documents and n_t is the number of documents containing term t .

Feature/ Technique	Description
Bag Of Words (BoW)	In this, the vector is crated using word frequency, irrespective of the word order
Term Frequency-Inverse Document Freqeunce (TF-IDF)	It measures and weighs the words in a document according to their significance
Vocabulary	It is simply all the unique words in a corpus, essential for creating vectors
Tokenization	Splitting the text into words or sub words so that the model can easily process them.

Table 1: Feature Extraction techniques

6.4 Model Development

At this stage, several machine learning as well as deep learning models shall be trained on the BBC News dataset for the purpose of news articles classification. In this regard the following statistical and probabilistic models shall be trained:

- Logistic Regression
- Support Vector Machines (SVC)
- Naïve Bayes and its variants
- Random Forest

These models shall be implemented on the vector or numerical representation of the textual data which shall be obtained using feature extraction techniques. These models are suitable for structured dataset.

In addition to these, neural network based deep learning models shall be designed and developed for training on this dataset. These models excel in capturing the contextual information as well as long term temporal dependencies in the sequential data such as text. Also, the attention mechanism shall be incorporated in these models which allows them to dynamically compute weightage of each token. Therefore these models are effective for modelling sequential data. The following deep learning model shall be tested and compared:

- LSTM
- GRU
- LSTM with attention layer
- BiLSTM
- BiLSTM with attention layer

The performance of all these models shall be computed and their results shall be compared and the best performing classifier shall be chosen for deploying in the unified system.

Along with these models, the transformer architecture based, decoder only LLMs such as Llama or GPT shall be employed which will help develop clear, concise, accurate and readable summaries of the fetched articles. This would be useful for abstractive summarization. Incorporation of such advanced models shall be very beneficial since it would help enhance the user experience as it would help the end user to gain insights about the lengthy news articles.

6.5 Real Time Pipeline

Subsequently, after training and evaluating various models, the best classifier shall be selected and it will be utilized for deployment purpose in real time news classification system. In this system, dynamic news articles shall be retrieved using an external news fetching APIs. Relevant information like title, description, and link of the articles shall be extracted. Also, a content extraction API shall also be used in order to fetch the entire article text.

This extracted text shall be further pre-processed and then passed to the trained model which will classify this article into a predefined category. Further, the full extracted news text shall be passed to the LLM for generating a clear summary. Through this way, the system will facilitate efficient news classification as well as news comprehension.

6.6 Architectural Flow Diagram

The architecture of the proposed IntelliNews Processing Framework is presented in Figure 1 below. The diagram offers a comprehensive structural view of the entire framework from collecting and preprocessing data to training a model, fetching real-time news, classifying them, and then using LLM to produce summaries of those articles in real time.

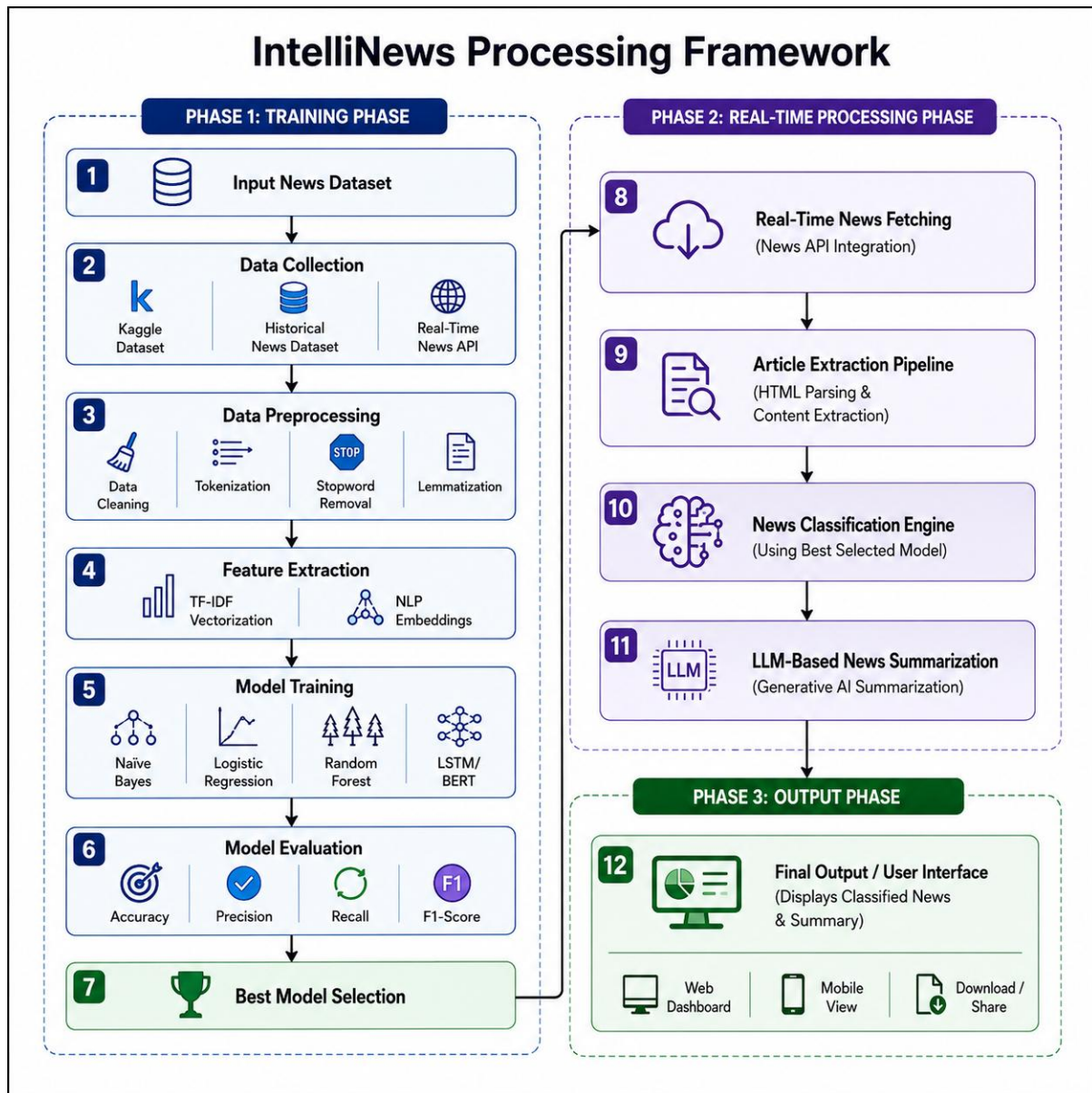


Figure 1: Flowchart of Proposed System

7 Dataset and Experimental Setup

This section includes the information about the dataset used in this research and the setup which is employed for the purpose of training as well as testing the classification models. It provides details about the characteristics of dataset, and strategies used for training models.

7.1 Dataset Description

The dataset which is used in this study is the BBC News Dataset which is sourced from BBC Website as well as Kaggle datasets. It consists of approximately 2,225 labelled news articles which are collected from BBC News portal. The dataset is in a structured format with rows and columns.

Property	Value
Dataset Name	BBC News Dataset
Total Samples	225
Number of Classes	5
Classes	Business, Sports, Technology, Entertainment, Politics
Data Type	Text (String)
Label Type	Categorical
Source	BBC Website, Kaggle

Table 2: Dataset Statistics

Each article text is associated to one of these pre-defined categories. Since this dataset has already labelled the corresponding classes for the articles, Therefore, it can be easily used for supervised learning tasks. The dataset is well balanced in terms of the number of classes and hence can be used for training machine learning and deep learning models without using class imbalance techniques.

7.2 Experimental Setup

All the experiments are carried out using the above mentioned BBC Dataset. The training and testing sets are created using the 80:20 ratio. This ensures that the models are trained on a vast variety of data and at the same time their remains sufficient unseen data for unbiased testing and evaluation of models predictions.

Additionally, these tests are conducted and run using a system with macOS operating system, and an Apple M1 chipset with an 8 GB RAM. All the classical models were trained using the CPU-powered system. However, the neural network based deep learning models were trained with the use of Google Colab, which is equipped with dedicated GPUs such as the NVIDIA T4 since these models require a much higher level of computation and parallel processing abilities which CPU alone cannot offer.

The experiments involves implementing and designing various machine learning models which include Logistic Regression, Naïve Bayes, Support Vector Machines, and Random Forest by using feature representations of the textual data that is obtained via Bag Of Words as well as TF-IDF techniques.

Also, deep learning or neural network based algorithms including LSTM, GRU, Bi-LSTM are implemented which addresses the issue of sequential as well as temporal dependencies in textual news articles. Furthermore, an attention mechanism has also been implemented along with these models which helps enable the model to focus on most significant and important words. Bi-LSTM model is also tested since the bi-directional layer allows the model to process text from both sides at the same time which helps it to understand the context more efficiently, sometimes leading to better results.

The primary language used for the purpose of this study is Python. Scikit-learn has been used for preprocessing the data, train-test splitting and implementing and designing the machine learning model and TensorFlow/Keras has been used in order to develop and train deep learning based models.

Also pandas, and NLTK frameworks have been used in order to convert raw data to a data frame and for standard pre-processing of the text respectively.

As for the ML models, standard configurations and hyperparameters have been followed, such as using linear and gaussian kernel for SVC and also using a constant number of estimators for random forest, to ensure a stable performance. Also, for the deep learning models, important hyperparameters, such as the batch size, number of epochs, embedding dimension size, and optimizer have been carefully chosen to ensure best performance. The task of the embedding layer is to convert the text into dense numeric vectors of fixed size which are able to capture the semantic meaning in the text. The number of epochs and batch size depend on balancing between the learning rate and the computational cost. Adam optimizer has been chosen as it was found to be the most effective and also for faster convergence.

The classification performance of these models are evaluated using several criteria as discussed in Section 6.8. Thus, based on the results obtained by each of these models, the best one is chosen for deployment.

Model	Configurations
Naïve Bayes	Multinomial NB, alpha = 0.1
SVC	C = 1.0, linear kernel, radial basis kernel
Random Forest	n_estimators = 100, max_depth = 20, min_samples_split = 5
Logistic Regression	C = 0.1, penalty = l2, solver = liblinear
LSTM	64 units, embedding_dim = 100, epochs = 30, batch_size = 32, Attention
GRU	64 units, embedding_dim = 100, epochs = 30, batch_size = 32
Bi-LSTM	64 units, embedding_dim = 100, Attention Layer, epochs = 30, Attention

Table 3: Model Configuration

7.3 Evaluation Metrics

The evaluation of the trained classification machine learning as well as neural network based models shall be evaluated using a variety of evaluation metrics. These metrics shall include Accuracy, Precision, Recall, F1-Score. These evaluation metrics shall give a holistic and overall understanding of the performance of the trained classification models. This would help in easily identifying the best performing which would be later used for deployment.

Metric	Description	Formula
Accuracy	It gives us a measure to determine overall effectiveness of the model	$TP + TN / (TP + TN + FP + FN)$
Precision	It tells us the number of positive instances that were correctly identified by classifier	$TP / TP + FP$
Recall	Tells number of positive instances that are correctly identified	$TP / TP + FN$
F1-Score	Determines the harmonic mean of the Precision as well as Recall	$(2 * P * R) / (P + R)$

Table 4: Performance Metrics used

TP, TN, FP, FN all are components of the confusion matrix which help in model evaluation.

8 Result and Analysis

The performance of all these trained models has been analysed and evaluated using accuracy, precision, recall and F1-score in order to find the optimal model for this task, with accuracy being the primary metric since the dataset was almost balanced. Apart from this, accuracy and loss graphs during training and validation of deep learning models were analysed.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Naïve Bayes (BoW)	97.64	97.60	97.7	97.62
Logistic Regression (BoW)	97.40	97.39	97.40	97.39
SVC (BoW)	97.40	97.39	97.40	95.39
Random Forest (BoW)	95.75	95.99	95.75	95.76
Naïve Bayes (TF-IDF)	95.75	96.07	95.43	95.66
Logistic Regression (TF-IDF)	97.641	97.648	97.641	97.643
SVC (TF-IDF)	98.34	98.35	98.34	98.34
Random Forest (TF-IDF)	96.22	96.42	96.226	96.224
LSTM	90.85	91.56	91.9	91.2
Bi-LSTM	94.60	95	94	94
Bi-LSTM (Self Attention)	95.31	96.0	95.2	95.1

Table 5: Performance comparison of results

Table 5. gives a good comparison of machine learning and neural network based models for news categorization. The Support Vector Classifier performed along with Bag of Word performed the best among all the models delivering an impressive accuracy of 98.34 % followed by Logistic Regression model. On the other hand, deep learning algorithm models such as Bi-LSTM and Bi-LSTM with attention layer offered an improvement in contextual learning.

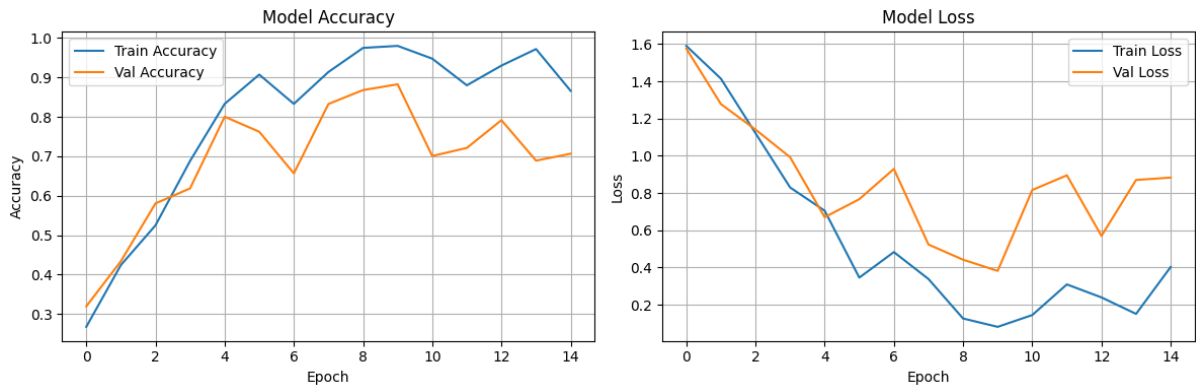


Figure 2: Tuned LSTM Model Performance

Figure 2. shows the training and validation loss and accuracy for the hyperparameter tuned LSTM model over many epochs. Training accuracy and validation accuracy increases over time but then becomes steady. On the other hand the training and validation loss decreases but the curve is not smooth, this may be due to certain parameters such as batch size or learning rate.

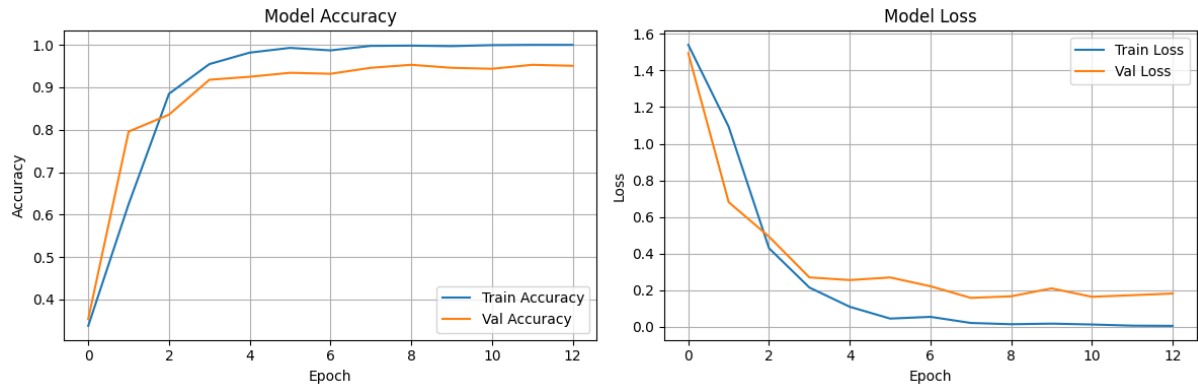


Figure 3: Tuned Bi-LSTM Model Performance

As can be observed from Figure 3, the performance of the optimized Bi-LSTM model in terms of accuracy and loss is observed during the training and testing phase. It performs better than the LSTM model in terms of validation accuracy. Loss for both the training and testing decreases steadily, which implies that it generalizes well without any overfitting issues.

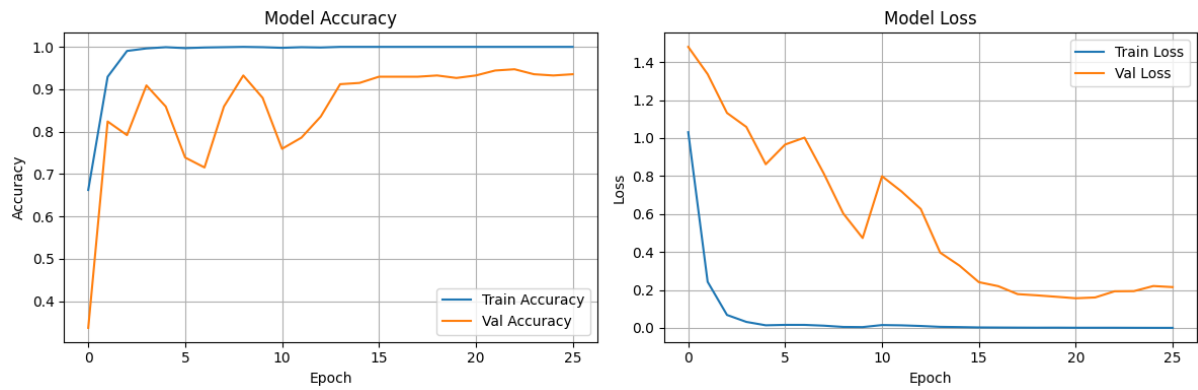


Figure 4: Bi-LSTM with Self Attention Model Performance

Figure 4 illustrates the performance and analysis of the Bi-LSTM model that has been developed along with self-attention layer. It can be observed that the graph and curves attains really good performance in terms of training accuracy and also manages to deliver good accuracy on the validation set as well. The validation loss also decreases steadily and there is considerable improvement in it when compared to the baseline LSTM model.

9 Conclusion

As can be seen and determined from the above results, the SVC model and TF-IDF combination got the best accuracy out of all the models that were used in this experiment. The TF-IDF technique, outperformed the Bag of Words technique in this case since it gives a higher weightage and value to significant and important words and also ignores frequently occurring words that maybe uninformative and not useful for the purpose of classification.

Also, the deep learning models such as LSTMs as well as Bi-LSTMs showed pretty good results in the experiments. This happened mainly because such models are very good at handling and analysing sequential data and temporal dependencies in the words. In addition, the incorporation of the Self Attention layer in the Bi-LSTM model significantly further improved its classification accuracy by giving higher weights to more significant words at each step and thereby enhancing contextual understanding.

At the same time it shall be observed that in this experiment, the classical and traditional machine learning models have performed better than advanced neural network based approaches. This may be due to the fact that the text articles which were utilized in the research had a less complex structure and may have contained important information and keywords which were directly associated with the predicted class and hence, it was not necessary to use sequential models to analyse context as they didn't help in improving the classification performance. Therefore, these classic algorithms were able to work more effectively here and give satisfactory results.

10 Limitations and Future Scope

Even though the models performed very well on the test data, and also in practice, there are however, still some downsides in these experiments. The first one being that the training and testing of all these models has been done using a single dataset. The usage of only this single dataset in the experiment would mean that the models exposure will be limited to only this dataset and it would miss varied writing styles, and complexities in the news articles. This may limit the models classification abilities as it may sometimes get complex articles.

Additionally, the framework as of now doesn't incorporate a human-in-the-loop feedback process which would enable and help the model to constantly learn from any corrections or feedbacks given to it via the users which would help it in adaptive refinement to enhance its classification capabilities.

One another drawback of the proposed system is that it relies only on one extraction API to fetch the article's text. This not only can increase the cost of the system, but it can also fail the system at the extraction point. If in case the API isn't working properly, it can affect the entire system.

Moreover, due to constraints in time and deadlines, the total integration of all the components, into a single unified framework was beyond reach. Although modules for each part have been developed separately, but effective as well as seamless combination of both the classification and summarization systems together in an end-to-end manner is still under consideration.

In future work, the system can be made more advanced and robust to handle real-world scenarios. This can be done in various ways including training the models on a vast variety of datasets and even developing our own training sets specifically for our use case. This would help the model capture a wide variety of nuances in the news articles which would help the model generalize well on unseen data and also enable it to easily classify complex articles in real time.

Furthermore, incorporating human-in-the-loop feedback mechanism could significantly improve the model's performance via the continuous refinement process. This would enable the system to receive human inputs about the predictions given via our model, and thus correcting any mistakes via continuous learning.

In future improvements, one great way of article extraction is via an intelligent agent-based approach that shall involve a combination of a few software utilities and selects the most suitable extraction method dynamically. For example, a free python web-scraping library such as Newspaper3k could be used as the first means of extracting the text from news articles. If at all there is any failure, such as entire text not being extracted, then in this case the extraction APIs can be automatically employed. It will be the duty of the agent to determine which tool to employ based on the output quality.

Additionally, this system can be enhanced to form a complete end-to-end solution that can carry out the task of news retrieval, categorization and also summarization via the trained machine learning, LLMs and the agentic based approach seamlessly and at scale. Advanced neural network based approaches such as transformer models like BERT can also be experimented with for procuring better results in terms of accuracy.

11 References

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